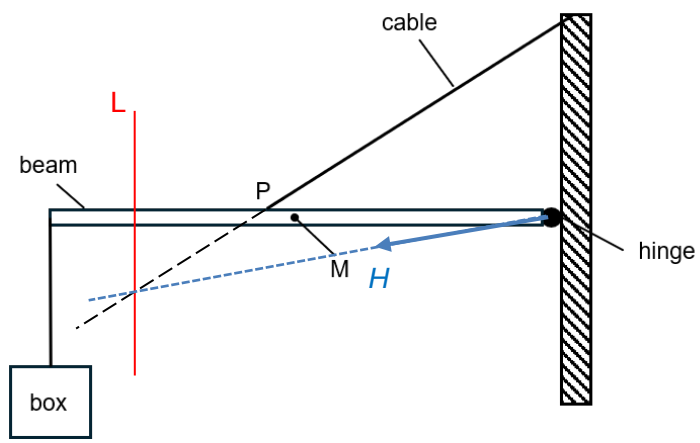
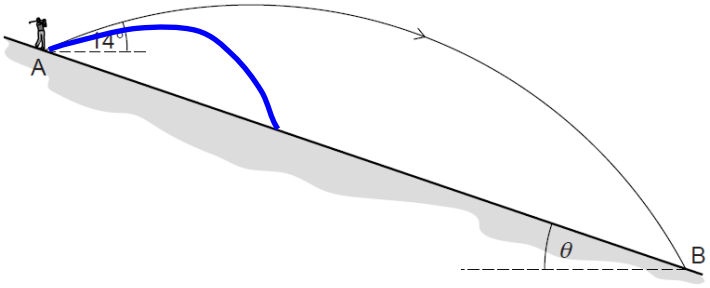
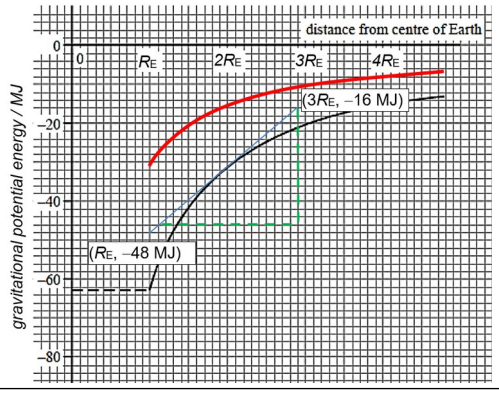
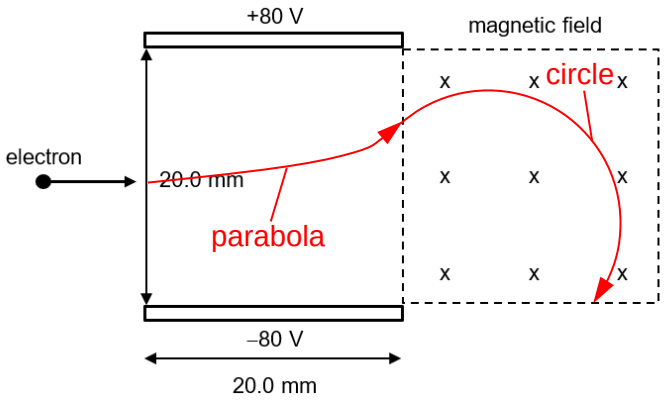
Annotations used in marking
BOD - Benefit of doubt
ECF - Error carried forward
POT - Powers of ten error
TE - Transfer error
CE - Calculation error
XP - Wrong physics
ENG - Generally bad english, phrasing and expression
PP - Poor presentation of answers

Qn	Suggested Answer
1	
(a)	1. Resultant force is zero.
	2. Resultant torque is zero about any axis.
(b)(i)1	
(b)(i)1	Correct line L drawn.
(b)(i)2	Correct force *H* drawn intersect with L and line of action of tension in cable.
(b)(ii)	The submerged box experiences an upthrust due to water. The downward force on the beam by the box is reduced.
	The line of action L will shift to the right.
	The angle that force *H* makes with the beam will decrease.
2	
(a)(i)	Taking upwards as positive, $s = ut + \frac{1}{2}at^2$ $s_V = (63\sin 14^\circ)(4.9) + \frac{1}{2}(-9.81)(4.9)^2$
	$s_V = -43.09$ m $= -43$ m
(a)(ii)	Consider horizontal component, $s_H = (63\cos 14^\circ)(4.9)$ $= 299.5$ m

	$\tan\theta = \dfrac{\lvert s_V \rvert}{\lvert s_H \rvert}$ $= \dfrac{43.09}{299.5}$ $\theta = 8.2^\circ$
(b)(i)	
(b)(ii)	Work is done against air resistance, gain in gravitational energy is lesser, hence maximum height is lower.
3	
(a)(i)	$v^2 = u^2 + 2as$ $85^2 = 120^2 + 2a(0.100)$ $a = -3.5875 \times 10^4$ $= -3.59 \times 10^4$ m s^{-2}
	deceleration = 3.59×10^4 m s^{-2}
(a)(ii)	$v = u + at$ $85 = 120 + (-3.5875 \times 10^4)t$ $t = 9.7561 \times 10^{-4}$ $= 9.8 \times 10^{-4}$ s
(b)(i)	There is no net external force or friction acting on the system.
(b)(ii)	$mu = (M + m)V$ $V = \left(\dfrac{m}{M+m}\right)u$ $= \left(\dfrac{0.40}{50.0 + 0.40}\right) \times 85$
	$= 0.6746$ $= 0.67$ m s^{-1}
(b)(iii)	Idea that total kinetic energy is conserved.
	Total initial kinetic energy of system $= \frac{1}{2}(0.40 \times 10^{-3})(85^2) = 1.445$ J Total final kinetic energy of system $= \frac{1}{2}(50.40 \times 10^{-3})(0.6746^2) = 0.0115$ J The collision is inelastic.
	OR
	Idea that relative speed of approach and separation is the same.
	Relative speed of approach = 85 m s^{-1} Relative speed of separation = 0 m s^{-1}

	The collision is inelastic.
4	
(a)(i)	Gravitational potential is defined as the work done per unit mass
	by an external agent in bringing a small test mass from infinity to that point, without producing any acceleration.
(a)(ii)	Accept gravitational potential between -62 and -64 MJ kg^{-1}
(b)(i)	Gravitational force provides centripetal force.
	$\frac{GMm}{R^2} = \frac{mv^2}{R}$
	$E_K = \frac{1}{2}mv^2$ $= \frac{GMm}{2R}$ (shown)
(b)(ii)	$E_T = E_K + U$
	$= \frac{GMm}{2R} + (-\frac{GMm}{R})$ $= -\frac{GMm}{2R}$
	
(c)(i)	$F_G = -\frac{dU}{dr}$
(c)(ii)	$F_G = -\frac{(-16-(-48))\times10^6}{3R_E - R_E}$ (given with tangent drawn on graph) $= -\frac{32\times10^6}{2\times6400\times10^3}$
	$= -2.5$ N
5	
(a)	e.m.f. is the amount of non-electrical energy converted into electrical energy per unit charge passing through the terminals of the cell (source).
	p.d. is the amount of electrical energy converted to other forms of energy per unit charge passing from one point to the other.
(b)(i)	p.d. across X = (9.0 – 4.0) = 5.0 V

	$R_X = \frac{5.0}{0.011}$ $= 450\ \Omega$
(b)(ii)1.	At 190 °C, R_Y = 25.0 Ω
	$V_Y = \left(\frac{R_Y}{(R_X + R_Y)}\right)(9.0)$ $= \left(\frac{25.0}{(450 + 25.0)}\right)(9.0)$
	$V_Y = 0.47$ V
b)(ii)2.	Ratio of R_Y to R_X will be lower. OR Total resistance increases.
	By the potential divider principle, the p.d. across Y will decrease. OR Current decreases, hence the p.d. across Y will decrease.
(c)(i)	p.d. across the 4.0 μF capacitor = 12 – 3.0 = 9.0 V
	$Q = CV$ $= (4.0 \times 10^{-6})(9.0)$
	$Q = 3.6 \times 10^{-5}$ C
(c)(ii)	Charge on 3.0 μF capacitor $= (3.0 \times 10^{-6})(3.0) = 9.0 \times 10^{-6}$ C
	Charge on Z $= 3.6 \times 10^{-5} - 9.0 \times 10^{-6} = 2.7 \times 10^{-5}$ C
	Capacitance of Z $= \frac{2.7 \times 10^{-5}}{3.0} = 9.0\ \mu F$
	OR
	Charge across 3.0 μF capacitor and Z = 3.6×10^{-5} C
	Effective capacitance of 3.0 μF capacitor and Z $= \frac{3.6 \times 10^{-5}}{3.0}$ $= 1.2 \times 10^{-5}$ F
	Capacitance of Z $= 1.2 \times 10^{-5} - 3.0 \times 10^{-6}$ $= 9.0 \times 10^{-6}$ $= 9.0\ \mu F$
6	

(a)	$v = \frac{s}{t}$ $t = \frac{s}{v}$ $= \frac{20.0 \times 10^{-3}}{9.0 \times 10^{6}}$ $= 2.22 \times 10^{-9}$ s (3 sf)
(b)	$F = qE$ $a = \frac{q\Delta V}{md}$ $= \frac{(1.60 \times 10^{-19})(80 + 80)}{(9.11 \times 10^{-31})(20.0 \times 10^{-3})}$ $= 1.41 \times 10^{15}$ m s^{-2}
	Taking upwards as positive, $v = u + at$ $= 0 + (1.41 \times 10^{15})(2.22 \times 10^{-9})$ $= 3.13 \times 10^{6}$ m s^{-1}
	Velocity $= \sqrt{v_x^2 + v_y^2}$ $= \sqrt{(9.0 \times 10^{6})^2 + (3.13 \times 10^{6})^2}$ $= 9.53 \times 10^{6}$ m s^{-1}
	$= 9.5 \times 10^{6}$ m s^{-1} (2 sf) (shown)
(c)	Magnetic force provides centripetal force. $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ $= \frac{(9.11 \times 10^{-31})(9.5 \times 10^{6})}{(0.72 \times 10^{-3})(1.60 \times 10^{-19})}$
	= 0.075 m (2 sf)
(d)	
	Upwards in a parabolic path between the parallel plates.
	Circular path in a clockwise direction within the magnetic field.

7	
(a)	Minimum C_D occurs at 27.0 m s^{-1} for the Jabulani.
(b)	The Jabulani has the narrowest and shallowest seams. It also has the shortest total seam length.
	Hence it has the lowest effective surface roughness.
	As Jabulani has the lowest C_D at the start, it will need a highest velocity for the drag force to reach a value where airflow around the ball transits from laminar flow to turbulent flow.
	Therefore, the smoother Jabulani requires the highest speed to reach the drag-crisis region.
(c)	From Fig. 7.2, $C_D = 0.15$ at 30 m s^{-1}
	$F_D = \frac{1}{2} C_D \rho A v^2$ $= \frac{1}{2}(0.15)(1.20)\pi\left(\frac{0.220}{2}\right)^2 (30^2)$ $= 3.08$ ≈ 3.1 N
(d)	For short passes between 12 and 15 m s^{-1}, Fig. 7.2 shows that the drag coefficient of the Jabulani slightly above 0.45 is greater than that of the Trionda around 0.2.
	For long kicks at speeds exceeding 25 m s^{-1}, Fig. 7.2 shows that the drag coefficient of the Trionda above 0.2 is greater than that of the Al Rihla and Jabulani below 0.2. Increased drag decelerates the ball more thus reducing the distance.
	For long kicks at speeds exceeding 25 m s^{-1}, Fig. 7.3 shows that the range of the Trionda is the lowest of the three balls.
(e)	rate of loss of kinetic energy, $L = F_D v$ $= \left(\frac{1}{2} C_D \rho A v^2\right) v$ $= \frac{1}{2} C_D \rho A v^3$ $\therefore L \propto C_D$
	From Fig. 7.2, C_D of Trionda = 0.22 at 30 m s^{-1}
	$\frac{L_J}{L_T} = \frac{C_{D,J}}{C_{D,T}}$ $= \frac{0.15}{0.22}$ $= 0.68$
(f)(i)	$\frac{F_D}{mg} = \frac{C_D \rho A v^2}{2mg}$ $= \frac{C_D}{2B}$

(f)(ii)	$\frac{F_D}{mg} = \frac{C_D \rho A v^2}{2mg}$ $= \frac{(0.22)(1.20)\pi\left(\frac{0.220}{2}\right)^2 (30)^2}{2(0.430)(9.81)}$ $=1.07$ ≈ 1.1
(f)(iii)	F_D is approximately equal to the weight of the ball. Hence, air resistance is not negligible.
(f)(iv)	Since $\frac{F_D}{mg} = \frac{C_D}{2B}$, when B tends to a large value, F_D is negligible compared to the weight of the ball.
	Hence, it will be a parabolic trajectory.
(g)(i)	At the top of the ball, the surface moves opposite to the airflow, reducing the airflow speed relative to the ball. Hence, the pressure at the top is higher than at the bottom. OR At the bottom of the ball, the surface moves in the same direction as the airflow, increasing the airflow speed relative to the ball. Hence, the pressure at the bottom is lower than at the top.
(g)(ii)	As speed is constant, kinetic energy remains constant. OR Magnus force is perpendicular to the displacement of the ball.
	Hence work done by Magnus force is zero.